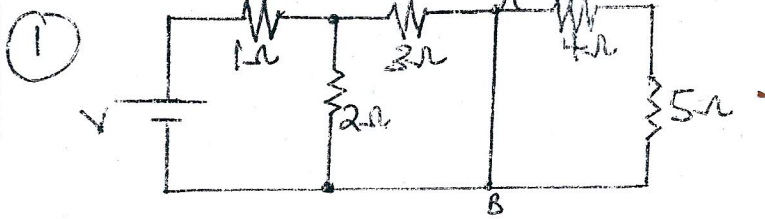


QASH V243723224

SOLUTION

SELF ASSESSMENT 1

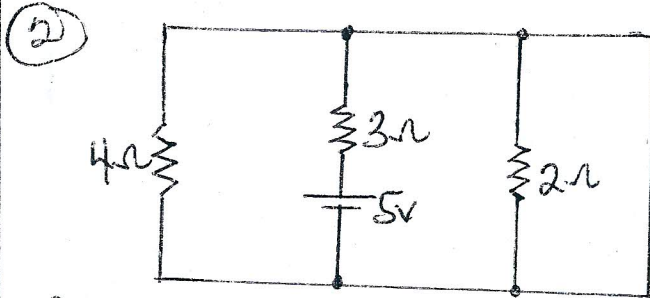


Path AB is a short circuit, hence current does not pass through 4Ω and 5Ω resistor.

$$\Rightarrow \text{Effective resistance } (R_T) = (2//3) + 1$$

$$= \frac{2 \times 3}{2 + 3} + 1$$

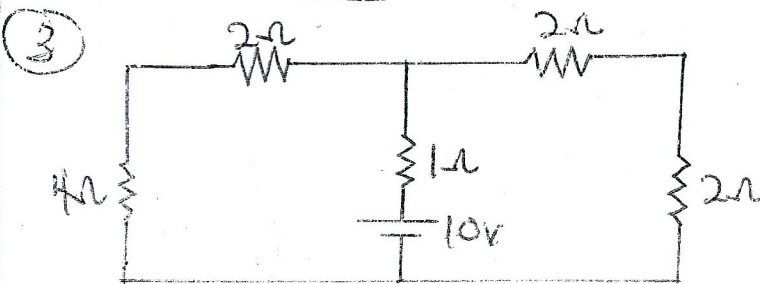
$$\therefore R_T = \underline{\underline{2.2\Omega}}$$



$$R_T = V = IR$$

$$I = \frac{V}{R} = \frac{5}{3}$$

$$\therefore I = \underline{\underline{1.67A}}$$



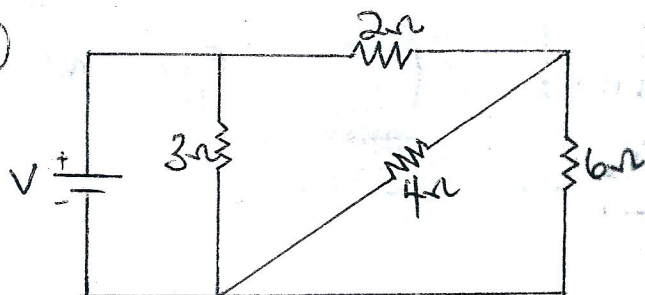
$$[(2+4)/(2+2)] + 1$$

$$(6//4) + 1$$

$$\frac{6 \times 4}{6 + 4} + 1 = 3.4\Omega$$

Effective resistance, $R_T = \underline{\underline{3.4\Omega}}$

4



$$[(4//6) + 2] // 3$$

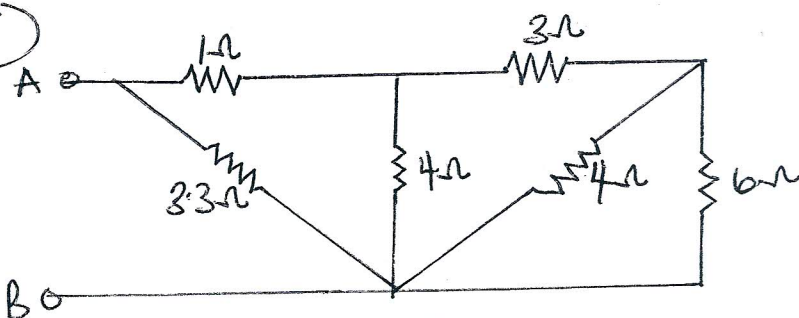
$$\left[\frac{4 \times 6}{4 + 6} + 2 \right] // 3$$

$$4.4 // 3$$

$$\frac{4.4 \times 3}{4.4 + 3} = 1.78 \Omega$$

$$\therefore R_T = \underline{1.78 \Omega}$$

5



$$\left[\left[\left[(4//6) + 3 \right] // 4 \right] + 1 \right] // 3.3$$

$$\Rightarrow \left[\frac{4 \times 6}{4 + 6} + 3 \right] // 4$$

$$\Rightarrow \left[5.4 // 4 \right] + 1$$

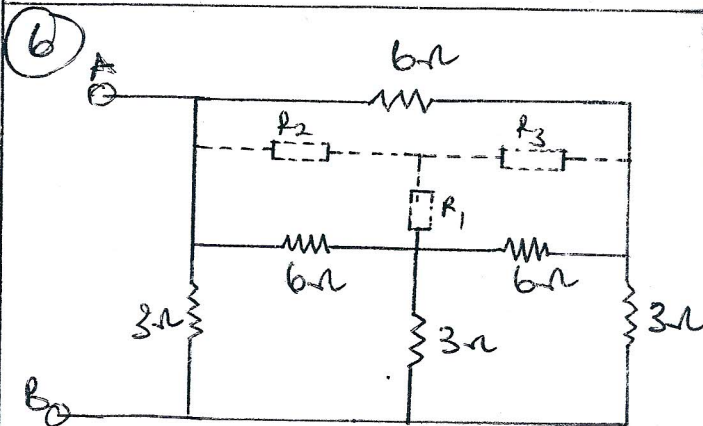
$$\frac{5.4 \times 4}{5.4 + 4} + 1$$

$$\left[\frac{15.5}{4.7} \right] // 3.3$$

$$\Rightarrow \frac{15.5}{4.7} \times 3.3$$

$$\frac{15.5}{4.7} + 3.3 = 1.649 \approx 1.65$$

$$\therefore \text{Effective resistance, } R_T = \underline{1.65 \Omega}$$

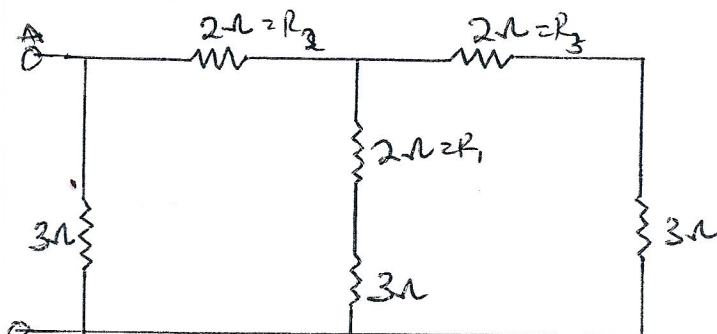


The three 6Ω -resistors are in ~~delta~~, so we change them to star.

Since all the three resistors have the same value, in changing them to star, we divide each by 3

$$\Rightarrow R_1 = \frac{6}{3} = 2\Omega \quad R_2 = \frac{6}{3} = 2\Omega \quad R_3 = \frac{6}{3} = 2\Omega$$

Transformed circuit



$$B \left[\left[(2+3) \parallel (2+3) \right] + 2 \right] \parallel 3$$

$$\Rightarrow \left[(5 \parallel 5) + 2 \right] \parallel 3$$

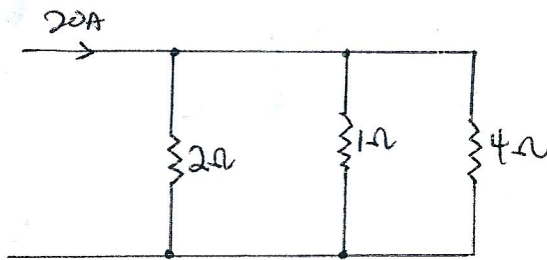
$$\Rightarrow \left[\frac{5 \times 5}{8+5} + 2 \right] \parallel 3$$

$$\Rightarrow 4.5 \parallel 3$$

$$\Rightarrow \frac{4.5 \times 3}{4.5 + 3} = 1.8\Omega$$

$$\therefore R_T = \underline{\underline{1.8\Omega}}$$

⑦



$$(2//4)$$

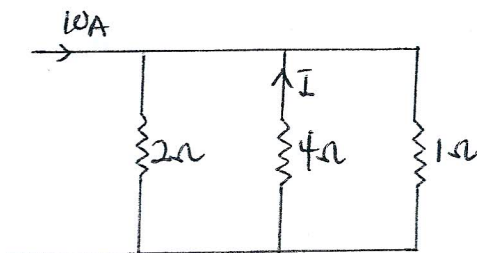
$$\Rightarrow \frac{2 \times 4}{2 + 4} = 1.333 \Omega$$

Using current division rule

$$I = \frac{1.333}{1.333 + 1} \times 20$$

$$\therefore I = \underline{\underline{11.43A}}$$

⑧



$$(2//1)$$

$$\Rightarrow \frac{2 \times 1}{2 + 1} = 0.6667$$

Using current division rule

$$I = \frac{0.6667}{0.6667 + 4} \times 10$$

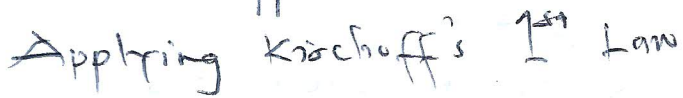
$$I = \underline{\underline{-1.429A}} \text{ (negative because } I \text{ is moving in opposite direction of source current).}$$

⑨

$$V = IR$$

$$\Rightarrow I = \frac{V}{R} = \frac{7}{5}$$

$$\therefore I = \underline{\underline{1.4A}}$$

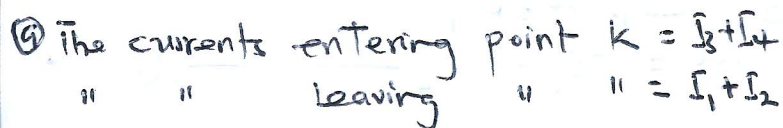


$$I = \underline{\underline{3A}}$$

$$V = L \frac{di}{dt} = 2 \times 3t \quad ; \quad \text{when } t = 4s$$

$$\Rightarrow V = 6(4) = \underline{\underline{24V}}$$

Given $I_2 = 3A$ $I_3 = 1A$ $I_4 = -2A$

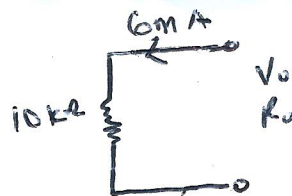


POWERED BY KUMI GEORGE

⑥

POWERED BY KUMI GEORGE

QUESTION 7



$$V_0 = (R_0 + 10 \times 10^3) 6 \times 10^{-3}$$

$$V_0 = 6 \times 10^{-3} R_0 + 60 \quad \text{--- (2)}$$

$$\Rightarrow (2) = (1)$$

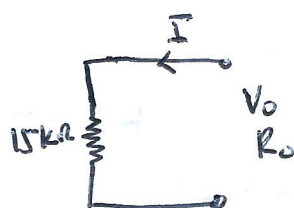
$$6 \times 10^{-3} R_0 + 60 = 10 \times 10^{-3} R_0$$

$$60 = 4 \times 10^{-3} R_0$$

$$R_0 = 15 \text{ k}\Omega$$

$$V_0 = 10 \times 10^{-3} \times 15 \times 10^3$$

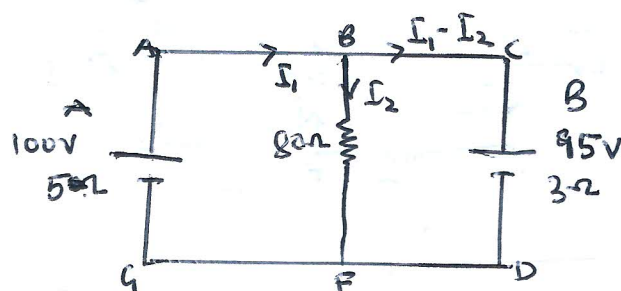
$$V_0 = 150 \text{ V}$$



$$V_0 = (R_0 + 15 \text{ k}\Omega) I$$

$$I = \frac{150}{15 \times 10^3 + 15 \times 10^3} = 5 \text{ mA}$$

QUESTION 9



⑨ Loop A B F G

$$100 = 5I_1 + 80I_2 \quad \text{--- (1)}$$

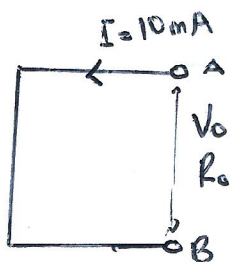
Loop B C D F

$$95 = -3(I_1 - I_2) + 80I_2$$

$$95 = 83I_2 - 3I_1 \quad \text{--- (2)}$$

$$I_1 = \frac{140}{131} \quad I_2 = \frac{155}{131} \text{ A}$$

QUESTION 8



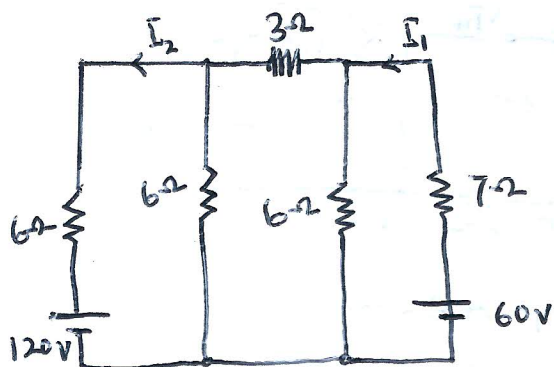
$$V_0 = I R_0$$

$$V_0 = 10 \times 10^{-3} R_0 \quad \text{--- (1)}$$

Current through bac. A = $I_1 = \frac{1.07A}{1.07} =$

(b) Current through $80\Omega = I_2 = 1.18A$

QUESTION 10



(a) When 120V source is deactivated

$$R_T = 6\Omega // 6\Omega + 3 // 6\Omega + 7\Omega$$

$$= 3 + 3 // 6 + 7$$

$$= 3 + 7 = 10\Omega$$

$$I_{1g} = \frac{60}{10} = 6A$$

(b) Current through 3Ω

$$\Rightarrow \frac{6}{6+6} \times 6(A) = 3A$$

$$\Rightarrow \bar{I}_2 = \frac{3}{2} = 1.5A //$$

(Since the resistors are the same)

(c) When 60V is deactivated

$$R_T = 7 // 6 + 3 // 6 + 6$$

$$= \frac{480}{53}\Omega$$

$$I_2 = \frac{120}{\frac{480}{53}} = 13.25A$$

Current through 3Ω

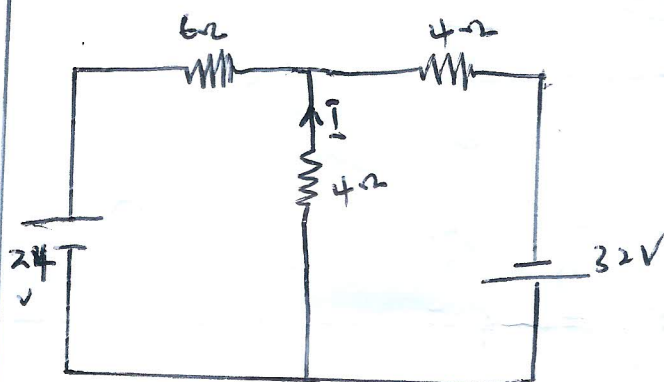
$$\frac{6}{6+6} \times 13.25 = 6.5$$

$$I_{1b} = \frac{6}{6+7} \times 6.5 = 3A$$

$$\Rightarrow I_1 = \cancel{6.5} I_{1g} - I_{1b}$$

$$= 6 - 3 = 3A$$

QUESTION 11



(a) If 24V bac. is acting alone

$$R_T = 4 // 4 + 6 = 8\Omega$$

$$I_g = \frac{24}{8} = 3A$$

(b) When 32V bac. is acting alone

$$R_T = 6 // 4 + 4 = \frac{32}{5}\Omega$$

$$I_2 = \frac{32 \times 5}{32} = 5A //$$

(c) Current through 4Ω when 24V is acting

$$I_a = \frac{4}{8} \times 3 = 1.5A$$

current through 4Ω when 32 is acting

$$I_b = \frac{6}{10} \times 5 = 3A$$

$$I = I_b - I_a = 3 - 1.5 = 1.5A$$

QUESTION 12.

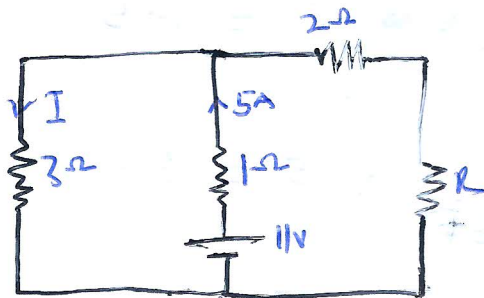


Fig. A

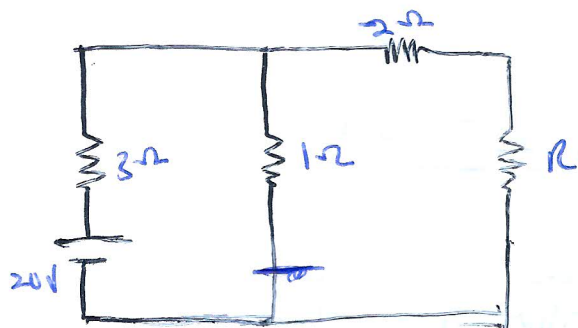


Fig. B

Current through 3Ω in fig A

$$11 = 5 \times 1 + 3I$$

$$I = \frac{6}{3} = 2A$$

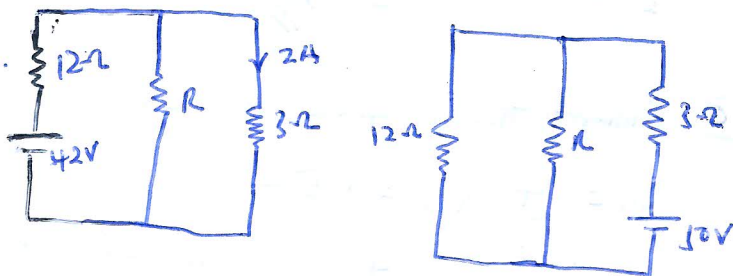
Using reciprocity theorem

$$I_f = 11 = 2A$$

$$20 = n$$

$$n = \frac{40}{11} = 3.636A$$

QUESTION 13



Finding current I through 3Ω when $42V$ and $2A$ are reciprocal

$$\Rightarrow 12(2) + 3I = 42$$

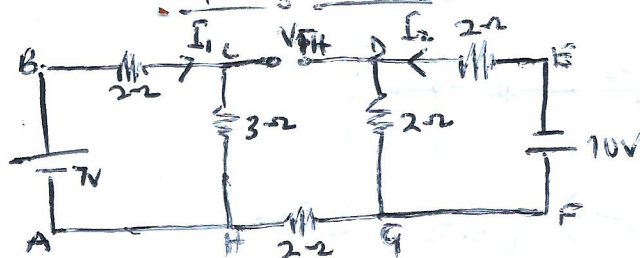
$$I = 6A$$

$$\text{Now } I_f = 42 = 6A$$

$$50 = x$$

$$\Rightarrow x = \frac{50 \times 6}{42} = 7.14A$$

QUESTION 14



(a) Loop ABCDA

$$7 = 2I_1 + 3I_1$$

$$I_1 = \frac{7}{5}A$$

Loop DEFGD

$$10 = 2I_2 + 2I_2$$

$$I_2 = -\frac{10}{4} = -\frac{5}{2}A$$

Loop CDGHC

$$V_{th} = 3I_1 - 2I_2$$

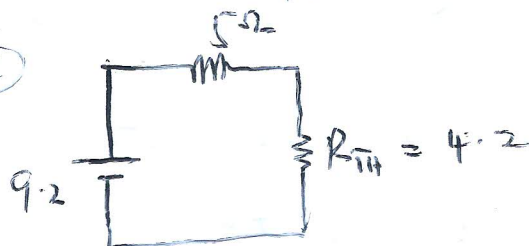
$$V_{th} = 3 \times \frac{7}{5} + 2 \times \frac{5}{2} = 9.2V$$

$$(b) 2 \parallel 3 + 2 \parallel 2 + 2 \parallel 2$$

$$\frac{6}{5} + 2 + 1 = 4.2V$$

$$R_{th} = 4.2\Omega$$

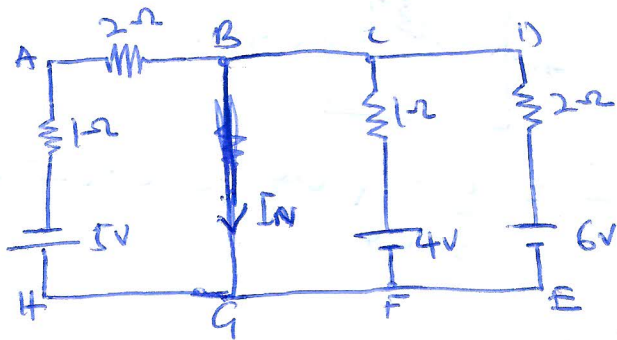
(c)



$$9.2 = I(5 + 4.2)$$

$$I = \frac{9.2}{9.2} = 1A$$

QUESTION 15



① Loop ABGHA

$$5 = -1I_N$$

$$I_{N1} = -5 \text{ A} \quad \text{①}$$

Loop BCFGB

$$4 = I_{N2} \quad \text{②}$$

Loop BCDEFGB

$$0 = 2I_{N3}$$

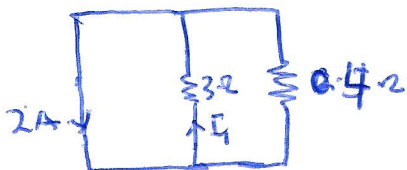
$$I_{N3} = 3 \text{ A} \quad \text{③}$$

$$I_N = I_{N1} + I_{N2} + I_{N3}$$

$$= -5 + 4 + 3 = 2 \text{ A} //$$

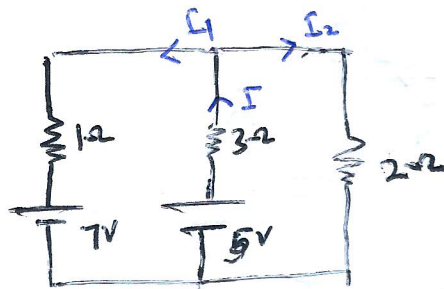
② $R_N = 1 \parallel 1 \parallel 2 \parallel 1 \parallel 2$

$$= \frac{0.5 \times 2}{0.5 + 2} = 0.4 \text{ } \cancel{2} \text{ } 0.4 \text{ } \Omega$$



$$I_1 = \frac{0.4}{3 + 0.4} \times 2 = 0.235 \text{ A} //$$

QUESTION 16



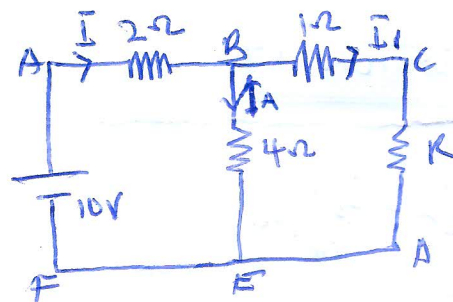
If 7V is deactivated

$$R_T = 1 \parallel 2 + 3 = \frac{2}{2} + 3 = \frac{11}{3} \text{ } \Omega$$

$$I = \frac{V}{R} = 5 \times \frac{3}{11} = \frac{15}{11} \text{ A}$$

$$I_2 = \frac{1}{3} \times \frac{15}{11} = 0.4545 \text{ A} //$$

QUESTION 17



Loop ABFEFA

$$10 = 2I + 4$$

$$I = 3 \text{ A} // \quad I_1 = 2 \text{ A} //$$

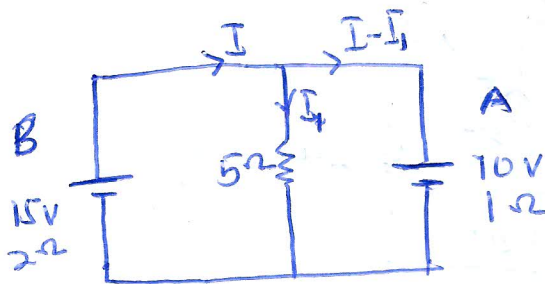
Loop BCDEB

$$0 = I_1 + I_2 R - 4$$

$$\Rightarrow 4 = 2 + 2R$$

$$R = 1 \text{ } \Omega //$$

QUESTION 18



$$\textcircled{a} \quad 15 = 2I + 5I_1 \quad \text{--- (1)}$$

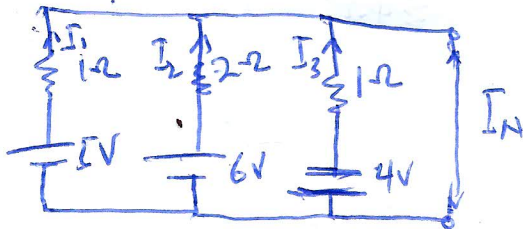
$$10 = -I + I_1 + 5I_1$$

$$10 = 6I_1 - I \quad \text{--- (2)}$$

$$I = 2.352 \text{ A}$$

$$\textcircled{b} \quad I_1 = 2.059 \text{ A}$$

QUESTION 19



$$I_1 = 5 \text{ A} \quad I_2 = 3 \text{ A} \quad I_3 = 4 \text{ A}$$

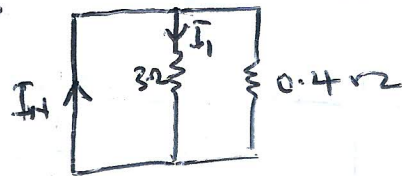
$$I_N = 5 + 3 - 4 = 4 \text{ A}$$

$$\textcircled{Q20.} \quad R_N = 1 \parallel 2 \parallel 1$$

$$= \frac{2}{3} \parallel 1$$

$$= \frac{2}{5} = 0.4 \Omega$$

Q21.



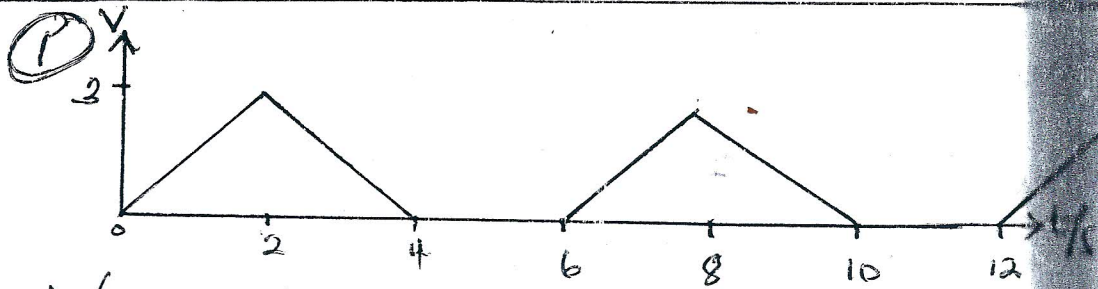
$$I_1 = \frac{0.4}{3 + 0.4} \times 0.4 = 0.047 \text{ A}$$

~~QUESTION 22 + 23~~

SOLUTIONS

Ref: MATEST 2019
KNUST

SELF ASSESSMENT 3



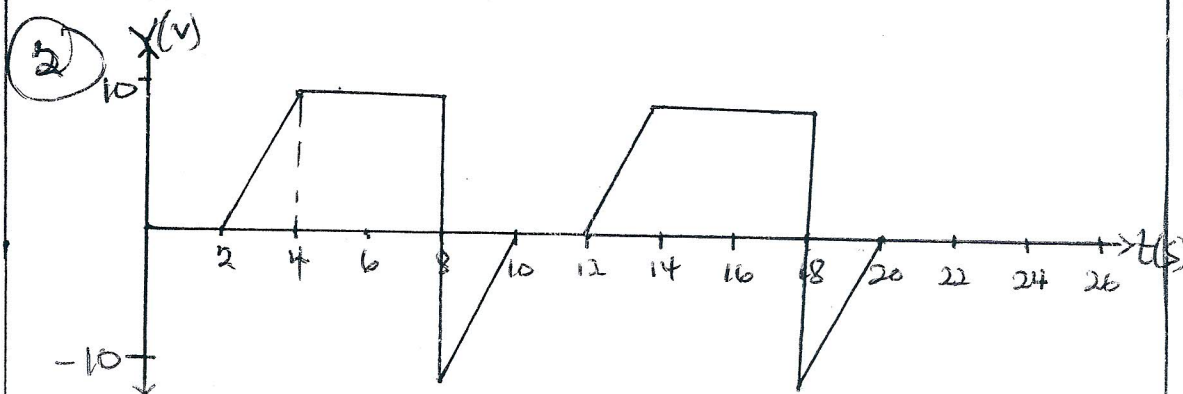
$$V_{rms} = \sqrt{\frac{(\text{area}(f))^2}{T}}$$

Squared area of right-angled triangle = $\frac{h^2 b}{3}$

$$V_{rms} = \sqrt{\frac{2\left(\frac{3^2 \times 2}{2}\right)}{6}}$$

$$V_{rms} = \sqrt{\frac{12}{6}} = \sqrt{2}$$

$$\therefore V_{rms} = \underline{\underline{1.414V}}$$



$$\textcircled{a} V_{avg} = \frac{\text{area}(f)t}{T}$$

$$= \frac{\left[\left(\frac{1}{2}bh\right) + (L \times b)\right] - \left(\frac{1}{2}bh\right)}{T}$$

$$\therefore V_{avg} = \left[\left(\frac{1}{2} \times 2 \times 10\right) + (10 \times 4)\right] - \left(\frac{1}{2} \times 2 \times 10\right)$$

DR-BRAH

$$② V_{rms} = \sqrt{\frac{\text{Area}(f^2 t)}{T}}$$

Area of Squared triangle = $\frac{h^2 b}{3}$

$$V_{rms} = \sqrt{\frac{\left[\left(\frac{10^2 \times 2}{3}\right) + (10^2 \times 4) + \left(\frac{(-10)^2 \times 2}{3}\right)\right]}{10}}$$

$$V_{rms} = \sqrt{\frac{\frac{200}{3} + 400 + \frac{200}{3}}{10}}$$

$$V_{rms} = \sqrt{\frac{160}{3}}$$

$$\therefore V_{rms} = \underline{\underline{7.3V}}$$

$$③ i(t) = 3 + 10 \sin(314t + 60^\circ) + 12 \sin(628t + 80^\circ) A$$

The wave current is a non-sinusoidal wave
Since the ~~frequency~~ frequencies are
not equal (i.e. $314t \neq 628t$)

$$\Rightarrow i_{rms} = \sqrt{i_0^2 + \left(\frac{i_m}{\sqrt{2}}\right)_1^2 + \left(\frac{i_m}{\sqrt{2}}\right)_2^2}$$

$$= \sqrt{3^2 + \left(\frac{10}{\sqrt{2}}\right)^2 + \left(\frac{12}{\sqrt{2}}\right)^2}$$

$$= \sqrt{131}$$

$$\therefore i_{rms} = \underline{\underline{11.446A}}$$

④ $V(t) = 2 + 15 \sin \omega t + 3 \sin(3\omega t + 16^\circ)$

The wave is a non-sinusoidal wave since the frequencies are not equal (i.e. $\omega t \neq 3\omega t$)

$$Z = 50 \Omega$$

$$I_{rms} = ?$$

$$V_{rms} = \sqrt{V_0^2 + \left(\frac{V_m}{\sqrt{2}}\right)_1^2 + \left(\frac{V_m}{\sqrt{2}}\right)_2^2}$$

$$V_{rms} = \sqrt{2^2 + \left(\frac{15}{\sqrt{2}}\right)^2 + \left(\frac{3}{\sqrt{2}}\right)^2}$$

$$\Rightarrow V_{rms} = 11V$$

$$\Rightarrow V_{rms} = I_{rms} Z$$

$$\Rightarrow I_{rms} = \frac{V_{rms}}{Z}$$

$$= \frac{11}{50}$$

$$\therefore I_{rms} = \underline{\underline{0.22A}}$$

⑤ $V(t) = 230 \sin \omega t = 230 \angle 0^\circ$

$$i(t) = 8 \cos \omega t + 8 \cos(\omega t + 160^\circ) + 6 \sin(\omega t + 30^\circ) + 2 \sin(\omega t - 30^\circ)$$

$$\text{but } 8 \cos \omega t = 8 \sin(\omega t + 90^\circ)$$

$$8 \cos(\omega t + 160^\circ) = 8 \sin(\omega t + 160^\circ + 90^\circ)$$

$$= 8 \sin(\omega t + 150^\circ)$$

Note: When changing cos to sin, add 90° to the angle

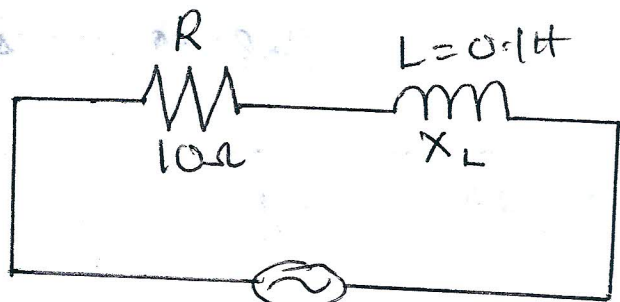
$$\begin{aligned} \Rightarrow i(t) &= 8 \sin(\omega t + 90^\circ) + 8 \sin(\omega t + 150^\circ) + 6 \sin(\omega t + 30^\circ) + 2 \sin(\omega t - 30^\circ) \\ &= 8 \angle 90^\circ + 8 \angle 150^\circ + 6 \angle 30^\circ + 2 \angle -30^\circ \\ &= 0 + j14 \end{aligned}$$

$$= 14 \angle 90^\circ$$

$$\therefore I_m = \underline{\underline{14}}$$

⑥ Phase angle between V and I is 90°

⑥



100V, 50Hz
 Impedance (Z) = $\sqrt{R^2 + X_L^2}$
 where $R = 10\Omega$ and $X_L = ?$
 but $X_L = \omega L$

$$= 2\pi fL$$

$$X_L = 2\pi \times 50 \times 0.1$$

$$X_L = 31.4\Omega$$

$$\Rightarrow Z = \sqrt{10^2 + 31.4^2}$$

$$\therefore Z = \underline{\underline{32.95\Omega}}$$

⑦

$$Z = 6 - j8 ; f_1 = 50\text{Hz}$$

$$Z = R - jX_C ; f_2 = 25\text{Hz}$$

$$X_{C1} = \frac{1}{2\pi f_1 C} \text{ and } X_{C2} = \frac{1}{2\pi f_2 C}$$

$$\Rightarrow \frac{X_{C2}}{X_{C1}} = \frac{1}{2\pi f_2 C} \div \frac{1}{2\pi f_1 C}$$

$$\Rightarrow \frac{X_{C2}}{X_{C1}} = \frac{2\pi f_1}{2\pi f_2}$$

$$X_{C2} = \frac{f_1}{f_2} \times X_{C1}$$

$$X_{C2} = \frac{50}{25} \times 8$$

$$X_{C2} = 16\Omega$$

For f_2 at 25Hz, its impedance is

$$Z = 6 - j16\Omega$$

DR-BRX

BT

⑧ Expressing in the form $a \pm jb$

$$\Rightarrow 25 \angle 30$$

$$\therefore 21.65 + j12.5$$

⑨ $V(t) = 230 \sin 314t$
 $i(t) = 10 \sin(314t - 30^\circ)$

Voltage and current can be rewritten as

$$V(t) = 230 \angle 0$$

$$i(t) = 10 \angle -30$$

$$\Rightarrow Z = \frac{V}{i} = \frac{230 \angle 0}{10 \angle -30}$$

$$\Rightarrow Z = 23 \angle 30, \text{ changing to rectangular form; hence,}$$

$$\Rightarrow Z = 19.919 + j11.5$$

① Resistance = $19.92 \underline{\underline{\Omega}}$

② Reactance (inductive) = $11.5 \underline{\underline{\Omega}}$

③ $P = \frac{V_m I_m}{2} \cos \phi$
 $= \frac{230 \times 10}{2} \cos 30$
 $= 1150 \cos 30$
 $\therefore P = 995.9 \approx 996 \underline{\underline{W}}$

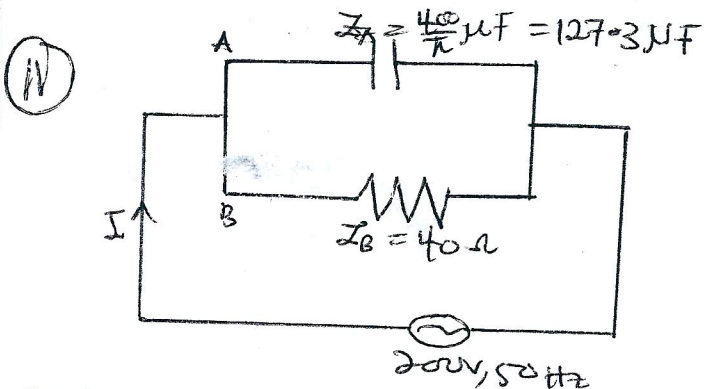
$$\textcircled{10} \quad Z_1 = 10 + j20 \Omega$$

$$Z_2 = -j12.5 \Omega$$

$$Z_T = Z_1 \parallel Z_2$$

$$Z_T = \frac{(10 + j20)(0 - j12.5)}{(10 + j20) + (0 - j12.5)}$$

$$Z_T = 10 - j20 \Omega$$



$$\textcircled{a} \quad Z_A = 0 - jX_C$$

$$X_C = \frac{1}{2\pi fC}$$

$$X_C = \frac{1}{2\pi \times 50 \times \frac{400}{\pi}} = 25 \mu F$$

$$Z_A = 0 - j25$$

$$Z_B = 40 + j0$$

$$Z_T = Z_A \parallel Z_B$$

$$Z_T = \frac{(-j25)(40)}{40 - j25}$$

$$Z_T = 11.235955 - j17.9775$$

$$Z_T = 21.2 \angle -58^\circ$$

$$\Rightarrow I = \frac{V}{Z_T} = \frac{200 \angle 0^\circ}{21.2 \angle -58^\circ}$$

$$\therefore I = 9.43 \angle 58^\circ$$

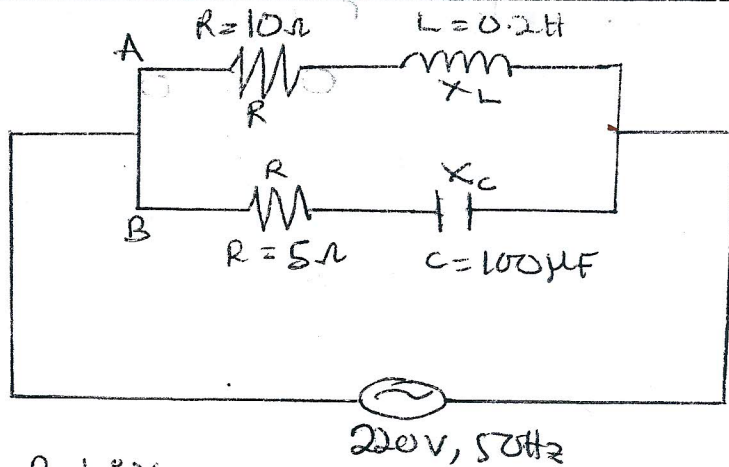
$$\textcircled{b} \quad P.f = \cos \phi$$

$$= \cos 58^\circ$$

$$P.f = 0.53$$

DR-READY
 BY

(12)



$$Z_A = R + jX_L$$

$$X_L = 2\pi fL$$

$$X_L = 2\pi \times 50 \times 0.2$$

$$X_L = 62.832 \Omega$$

$$Z_A = 10 + j62.832 \Omega$$

$$Z_B = R + jX_C$$

$$X_C = \frac{1}{2\pi fC}$$

$$X_C = \frac{1}{2\pi \times 50 \times 100}$$

$$X_C = 31.831 \Omega$$

$$Z_B = 5 - j31.831 \Omega$$

$$Z_T = Z_A \parallel Z_B$$

$$Z_T = \frac{(10 + j62.832)(5 - j31.831)}{(10 + j62.832)(5 - j31.831)}$$

$$Z_T = 59.524 \angle -64.296^\circ$$

$$\therefore Z_{T(p)} = 59.524 \Omega$$

$$(B) \ I = \frac{V}{Z_T}$$

$$= \frac{220 \angle 0^\circ}{59.524 \angle -64.296^\circ}$$

$$I = 1.603 + j3.33$$

$$I_T = 3.696 \angle 64.296^\circ$$

$$\Rightarrow I_A = \frac{V_A}{Z_A} = \frac{220 \angle 0^\circ}{63.622796 \angle 80.957^\circ}$$

$$I_A = 0.543 - j3.415$$

$$= 3.457 \angle -80.957^\circ$$

$$\therefore I_A = 3.457 A$$

$$(C) \ P.f = \cos \phi$$

$$(C) \ R.C \text{ part} = Z_B$$

$$Z_B = 5 - j31.831$$

$$Z_B = 32.221 \angle -81.07^\circ$$

$$\Rightarrow P.f = \cos \phi$$

$$= \cos -81.07^\circ$$

$$\therefore P.f = 0.16 \text{ leading}$$

DR-BRAIN

13

Load (S) (KVA)	P.f ($\cos \phi$)	P (kW)	$\sin \phi$	Q (VAR)
(i) 1.5	0.8	1.2	0.6	0.9
(ii) 1	1	1	0	0

$$S = \frac{P}{\cos \phi} = \frac{1}{1} = 1 \text{ KVA}$$

$$P = S \cos \phi$$

$$= 1.5 \times 0.8 = 1.2$$

$$\sin^2 \phi + \cos^2 \phi = 1$$

$$\sin^2 \phi = 1 - \cos^2 \phi$$

$$\sin \phi = \sqrt{1 - \cos^2 \phi}$$

$$\sin \phi = \sqrt{1 - P_f^2} = \sqrt{1 - (0.8)^2} = 0.6 \text{ for when } P.f = 0.8$$

Again, when $P.f = 1$

$$\sin \phi = \sqrt{1 - 1^2} = 0$$

$$Q = S \sin \phi \text{ when } S = 1.5$$

$$Q = 1.5 \times 0.6 = 0.9$$

when $S = 1$

$$Q = 1 \times 0 = 0$$

(a) Total power = $1.2 + 1$

$$= 2.2 \text{ kW}$$

(b) Total reactive power = $0.9 + 0$

$$= 0.9 \text{ VAR}$$

(c) $P = 0.8$ and $Q = 0.6$

$$S = \sqrt{P^2 + Q^2}$$

$$S = \sqrt{0.8^2 + 0.6^2} = 1 \text{ KVA}$$

$$\Rightarrow S = IV$$

$$I = \frac{S}{V}$$

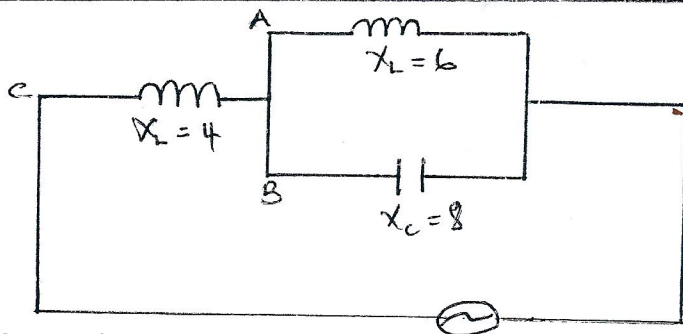
$$I = \frac{1 \text{ KVA}}{200} = \frac{1000}{200}$$

$$\therefore I = 5 \text{ A}$$

DR-BR

BY

(14)



$$Z_L = j4 \Omega \quad Z_A = j6 \Omega \quad Z_C = -j8 \Omega$$

$$Z_T = Z_L + (Z_A \parallel Z_C)$$

$$= j4 + \frac{(j6)(-j8)}{j6 + (-j8)}$$

$$= j4 + j24$$

$$Z_T = j28 \Omega$$

$$\therefore Z_T = 28 \angle 90^\circ$$

$$(15) \quad i = (2 - j2) A$$

$$V = (100 - j150) V$$

$$S = V i^* = P + jQ$$

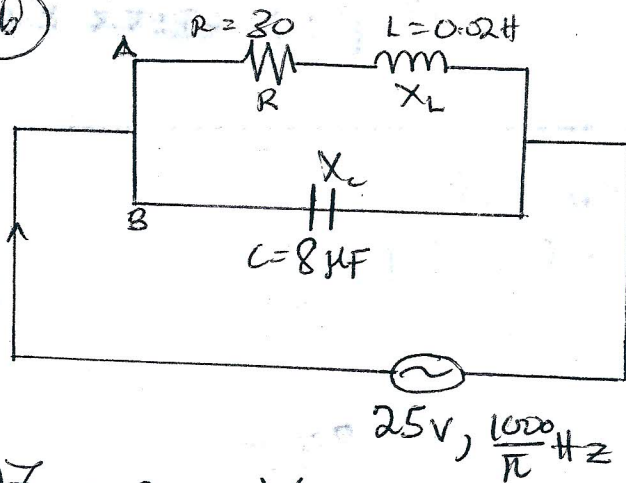
$$S = (100 - j150)(2 + j2)$$

where i^* is the conjugate of i

$$\Rightarrow S = 500 - j100$$

$\therefore$ Reactive power, $Q = 100 \text{ VAR}$ (capacitive)
Since the reactive part is negative.

16)



$$\begin{aligned} Z_A &= 30 + jX_L \\ &= 30 + j2\pi fL \\ &= 30 + j2\pi \times \frac{1000}{\pi} \times 0.02 \end{aligned}$$

$$Z_A = 30 + j40 \Omega$$

$$Z_B = 0 - jX_C$$

$$= -j \frac{1}{2\pi fC}$$

$$= -j \frac{1}{2\pi \times 1000 \times 8 \times 10^{-6}}$$

$$Z_B = -j62.5$$

$$Z_T = Z_A \parallel Z_B$$

$$= \frac{(30 + j40)(-j62.5)}{(30 + j40) + (-j62.5)}$$

$$Z_T = 83.333 \Omega$$

$$\Rightarrow I = \frac{V}{Z_T}$$

$$I = \frac{25 \angle 0^\circ}{83.33 \angle 0^\circ}$$

$$\therefore I = 0.3 \text{ A}$$

⑥ $P = IV$

$$= 0.3 \times 25$$

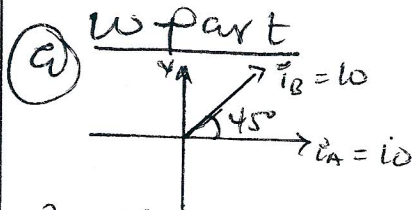
$$\therefore P = 7.5 \text{ W}$$

BY DR. SRINIVAS

$$(17) \quad i_A = 10 \sin \omega t + 5 \sin 3\omega t$$

$$i_B = 10 \sin(\omega t + \frac{\pi}{4}) + 5 \sin(3\omega t - \frac{\pi}{6})$$

$$I_T = i_A + i_B$$



Resolving x-component

$$10 \cos 45 + 10 \cos 0$$

$$7.071 + 10$$

$$= 17.071$$

Resolving y-component

$$10 \sin 45 + 10 \sin 0$$

$$7.071 + 0$$

$$= 7.071$$

$$\text{Amplitude, } I_m = \sqrt{17.071^2 + 7.071^2}$$

$$= 18.4775 \text{ A}$$

$$\phi = \tan^{-1} \left(\frac{7.071}{17.071} \right) = 22.5^\circ$$

$$\text{Hence } 18.4775 \sin(\omega t + 22.5^\circ)$$

$$\text{Therefore } i_T = 18.4775 \sin(\omega t + 22.5^\circ) + 9.659 \sin(3\omega t - 15^\circ)$$

$$I_{rms} = \sqrt{i_1^2 + \left(\frac{I_m}{\sqrt{2}} \right)_1^2 + \left(\frac{I_m}{\sqrt{2}} \right)_2^2}$$

$$\therefore I_{rms} = \sqrt{0^2 + \left(\frac{18.4775}{\sqrt{2}} \right)^2 + \left(\frac{9.659}{\sqrt{2}} \right)^2}$$

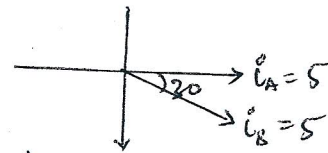
$$= 14.743 \text{ A}$$

$$(B) \quad Z_T = \frac{V_{rms}}{I_{rms}}$$

$$Z_T = \frac{200}{14.743}$$

$$= 13.566 \Omega$$

3ω - part



Resolving x-component

$$5 \cos 0 + 5 \cos 30$$

$$5 + 4.33$$

$$= 9.33$$

Resolving y-component

$$5 \sin 0 - 5 \sin 30$$

$$0 - 2.5$$

$$= -2.5$$

$$\text{Amplitude, } I_m = \sqrt{2.5^2 + 9.33^2}$$

$$= 9.659 \text{ A}$$

$$\phi = \tan^{-1} \left(\frac{-2.5}{9.33} \right)$$

$$\phi = -15^\circ$$

$$\text{Hence } 9.659 \sin(3\omega t - 15^\circ)$$

18

Load (S) (kVA)	Pf (cos ϕ)	P (kW)	sin ϕ	Q (Var)
(i) 2.5 kVA	0.85	2.125	0.5268	1.317
(ii) 1.5 kVA	0.90	1.35	0.436	- 0.654

$$\begin{aligned} \text{(i)} \quad P &= S \cos \phi \\ &= 2.5 \times 0.85 \\ &= 2.125 \text{ kW} \end{aligned}$$

$$\begin{aligned} \sin \phi &= \sqrt{1 - \text{pf}^2} \\ &= \sqrt{1 - 0.85^2} \\ &= 0.5268 \end{aligned}$$

$$\begin{aligned} Q &= S \sin \phi \\ &= 2.5 \times 0.5268 \\ &= 1.317 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P &= S \cos \phi \\ &= 1.5 \times 0.9 \\ &= 1.35 \text{ kW} \end{aligned}$$

$$\begin{aligned} \sin \phi &= \sqrt{1 - \text{pf}^2} \\ &= \sqrt{1 - 0.9^2} \\ &= 0.436 \end{aligned}$$

$$\begin{aligned} Q &= S \sin \phi \\ &= 1.5 \times 0.436 \\ &= 0.654 \end{aligned}$$

NOTE: In the above problem, Since one load is having a leading power factor, then its Q would have a negative sign and thus would have to be subtracted from the others.

$$\begin{aligned} \text{(a)} \quad S_T &= \sqrt{P_T^2 + Q_T^2} \\ P_T &= 2.125 + 1.35 \\ &= 3.475 \text{ kW} \\ Q_T &= 1.317 - 0.654 \\ &= 0.663 \text{ VAR} \end{aligned}$$

$$\begin{aligned} S_T &= \sqrt{3.475^2 + 0.663^2} \\ \therefore S_T &= \underline{\underline{3.538 \text{ kVA}}} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \cos \phi &= \frac{P_T}{S_T} \\ &= \frac{3.475}{3.538} \\ \therefore \cos \phi &= \underline{\underline{0.98 \text{ lagging}}} \end{aligned}$$

DR. B. R. K.

$$(19) V = 200 ; f = 50 \text{ Hz}$$

$$i = 5 \sin(314t - \frac{\pi}{3}) \text{ A}$$

$$Z = \frac{V_{\text{rms}}}{I_{\text{rms}}}$$

$$I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$$

$$I_{\text{rms}} = \frac{5}{\sqrt{2}} = 3.5355 \text{ A}$$

$$Z = \frac{200}{3.5355} = 56.569 \Omega$$

$$(20) X_L = Z \sin \phi$$

$$X_L = 56.569 \sin 60$$

$$X_L = 48.9902$$

$$\text{NB: } \frac{\pi}{3} = \frac{180}{3} = 60^\circ$$

$$X_L = 2\pi fL$$

$$L = \frac{X_L}{2\pi f} = \frac{48.9902}{2\pi \times 50}$$

$$L = 0.15594 \text{ H}$$

$$\therefore L = \underline{\underline{155.9 \text{ mH}}}$$

$$(6) R = Z \cos \phi$$

$$R = 56.569 \cos 60$$

$$\therefore R = \underline{\underline{28.285 \Omega}}$$

$$(20) \quad V = 200 + j10$$

$$I = 2 + j5$$

$$S = VI^* = P + jQ$$

$$S = (200 + j10)(2 - j5)$$

$$S = 480 - j980 = P + jQ$$

$$P = 480 \text{ W}$$

$\therefore$ Reactive power supplied, $Q = 980 \text{ VAR}$.

$$(21) \quad P = 2 \text{ kW}$$

$$\eta = 80\% = 0.8$$

$$\text{PF}(\cos\phi) = 0.75 \text{ lagging}$$

$$\eta = \frac{\text{out put power}}{\text{input power}}$$

$$\text{Input power} = \frac{\text{output power}}{\eta}$$

$$\text{Input power} = \frac{2000}{0.8} = 2500 \text{ W}$$

$$S = \frac{\text{out put power}}{\eta \times \text{PF}}$$

$$S = \frac{2000}{0.8 \times 0.75}$$

$$S = 3333.33 \text{ VA} = 3.33 \text{ kVA}$$

$$S = IV$$

$$\rightarrow I = \frac{S}{V} = \frac{3333.33}{240}$$

$$\therefore I = 13.889 \text{ A}$$

$$(b) \quad Q = S \sin\phi$$

$$\sin\phi = \sqrt{1 - \text{PF}^2} = \sqrt{1 - 0.75^2} = 0.6614$$

$$\rightarrow Q = 3.3 \text{ kVA} \times 0.6614$$

$$\therefore Q = 2.205 \text{ kVAR}$$

SOLUTION

SELF ASSESSMENT 4

Ref: MATISA KNUST

$$\textcircled{a} Z_s = 8 + j6$$

$$= 10 \angle 36.87^\circ$$

NB: Z_s = Star impedance

Z_{se} = equivalent star impedance.

$$Z_{se} = \frac{24 + j18}{3}$$

$$= 10 \angle 36.87^\circ$$

$$V_L = 240\sqrt{3} ; V_p = 240V$$

$$I_L = I_p$$

$$V_p = I_p Z_s$$

$$I_{ps} = \frac{V_p}{Z_s} = \frac{240 \angle 0^\circ}{10 \angle 36.87^\circ}$$

$$I_{ps} = 24 \angle -36.87^\circ$$

$$I_{pze} = \frac{V_p}{Z_{se}} = \frac{240 \angle 0^\circ}{10 \angle 36.87^\circ} = 24 \angle -36.87^\circ$$

$$I_p = I_{ps} + I_{pze}$$

$$= 24 \angle -36.87^\circ + 24 \angle -36.87^\circ$$

$$I_p = 48A$$

$$\therefore I_p = I_L = \underline{48A}$$

$$\textcircled{b} Z_d = 24 + j18 \Omega = 30 \angle 36.87^\circ$$

$$I_L = I_p \sqrt{3}$$

$$I_p = \frac{I_L}{\sqrt{3}}$$

$$V_p = V_L$$

$$V_L = I_L Z$$

$$I_L = \frac{V_L}{Z} = \frac{240\sqrt{3} \angle 0^\circ}{30 \angle 36.87^\circ} = 13.856 \angle -36.87^\circ$$

$$\Rightarrow V_L = 13.856 \angle -36.87^\circ \times 30 \angle 36.87^\circ$$

$$V_L = 415.68V$$

$$V_p = I_p Z$$

$$I_p = \frac{V_p}{Z} = \frac{415.68}{30}$$

$$\therefore I_p = \underline{13.85A}$$

$$\begin{aligned} \textcircled{2} \quad V_L &= 240\sqrt{3} \\ f &= 50 \text{ Hz} \\ S &= 3.6 \text{ kVA} \\ \text{Pf} (\cos \phi) &= 0.6 \text{ lagging} \end{aligned}$$

$$\begin{aligned} \textcircled{a} \quad S &= \sqrt{3} V_L I_L \\ I_L &= \frac{S}{\sqrt{3} V_L} \\ I_L &= \frac{3.6 \times 10^3}{\sqrt{3} \times 240\sqrt{3}} \\ \therefore I_L &= \underline{\underline{5 \text{ A}}} \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad Z &= \frac{V_p}{I_p} \\ Z &= \frac{240}{5} = 48 \Omega \\ \cos \phi &= 0.6 \\ \phi &= \cos^{-1} 0.6 \\ \phi &= 53.13^\circ \\ Z &= 48 < 53.13^\circ \\ \text{In the form } Z &= R + jX_L \\ Z &= 28.8 + j38.4 \\ \therefore \text{Reactance, } X_L &= \underline{\underline{38.4 \Omega}} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad I_N &= I_{p_R} + I_{p_Y} + I_{p_B} \\ \textcircled{a} \quad I_N &= 40 \angle 0^\circ + 50 \angle -120^\circ + 60 \angle 120^\circ \\ I_N &= -15 + j5\sqrt{3} \\ I_N &= 10\sqrt{3} \angle 150^\circ \\ \Rightarrow I_N &= 17.3 \angle 150^\circ \\ \therefore I_N &= \underline{\underline{17.3 \text{ A}}} \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad P_R &= V_p I_p \cos \phi \\ &= 230 \times 40 \times 0.9 \\ &= 8280 \text{ W} \\ P_Y &= 230 \times 50 \times 0.9 \\ &= 10350 \text{ W} \\ P_B &= 230 \times 60 \times 0.9 \\ &= 12420 \text{ W} \end{aligned}$$

$$\begin{aligned} \Rightarrow P_T &= P_R + P_Y + P_B \\ P_T &= 8280 + 10350 + 12420 \\ P_T &= 31050 \text{ W} \\ \therefore P_T &= \underline{\underline{31.05 \text{ kW}}} \end{aligned}$$

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Ref: MATISA KNUST

$$\textcircled{A} \quad V_L = 250\sqrt{3} \\ f = 50 \text{ Hz}$$

$$S = 3.75 \text{ kVA} \\ \text{pf} (\cos \phi) = 0.6 \text{ lagging}$$

$$\textcircled{B} \quad S = \sqrt{3} V_L I_L$$

$$I_L = \frac{S}{\sqrt{3} V_L}$$

$$\Rightarrow I_L = \frac{3.75 \times 10^3}{\sqrt{3} \times 250\sqrt{3}}$$

$$\therefore I_L = \underline{\underline{5 \text{ A}}}$$

$$\textcircled{C} \quad X_L = 2\pi f L$$

$$L = \frac{X_L}{2\pi f}$$

$$Z = \frac{V_p}{I_p} = \frac{250}{5} = 50 \Omega$$

$$\cos \phi = 0.6$$

$$\phi = \cos^{-1} 0.6$$

$$\phi = 53.13$$

$$\Rightarrow Z = 50 \angle 53.13$$

In the form $Z = R + jX_L$

$$Z = 30 + j40 \Omega$$

$$\Rightarrow X_L = 40 \Omega$$

$$\Rightarrow L = \frac{40}{2\pi \times 50}$$

$$\therefore L = \underline{\underline{0.127 \text{ H}}}$$

BY ~~DR. B. R. R. R.~~

(5) $V_L = 400$

a) $P_P = V_P I_P$

$I_P = \frac{P_P}{V_P}$

for star-connected

$I_P = I_L$

$V_L = \sqrt{3} V_P$

$V_P = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$

$I_{P_{5kW}} = \frac{5 \times 10^3}{230.94} = 21.65 \text{ A}$

$\therefore I_{P_{5kW}} = \underline{\underline{21.65 \text{ A}}}$

(8) $I_N = I_{P_{5kW}} + I_{P_{4kW}} + I_{P_{2.5kW}}$

$I_L = I_{P_{4kW}} = \frac{P_P}{V_P} = \frac{4 \times 10^3}{230.94} = 17.3205 \text{ A}$

$I_L = I_{P_{2.5kW}} = \frac{P_P}{V_P} = \frac{2.5 \times 10^3}{230.94} = 10.8253 \text{ A}$

$\Rightarrow I_N = 21.65 \angle 0^\circ + 17.3205 \angle -120^\circ + 10.8253 \angle 120^\circ$

$I_N = 7.575 - j5.6205$

$= 9.43 \angle -36.57^\circ$

$\therefore I_N = \underline{\underline{9.43 \text{ A}}}$

(6) $R = 5 \Omega$ $L = 66/\pi \text{ mH}$

balanced star connected

$V_L = 250\sqrt{3}$ $f = 50 \text{ Hz}$

(a) $I_L = I_P$

$X_L = \omega L = 2\pi fL$

$X_L = 2\pi \times 50 \times 66/\pi \times 10^{-3}$

$X_L = 6.5998 \Omega$

$Z = 5 + j6.5998$

$V_P = I_P Z$

$V_L = \sqrt{3} V_P$

$\Rightarrow V_P = \frac{V_L}{\sqrt{3}} = \frac{250\sqrt{3}}{\sqrt{3}} = 250 \text{ V}$

$I_P = \frac{V_P}{Z} = \frac{250 \angle 0^\circ}{5 + j6.5998}$

$I_L = I_P = 30.192 \angle -52.81^\circ$

$\therefore I_L = \underline{\underline{30.2 \text{ A}}}$

(b) $P = \sqrt{3} V_L I_L \cos \phi$

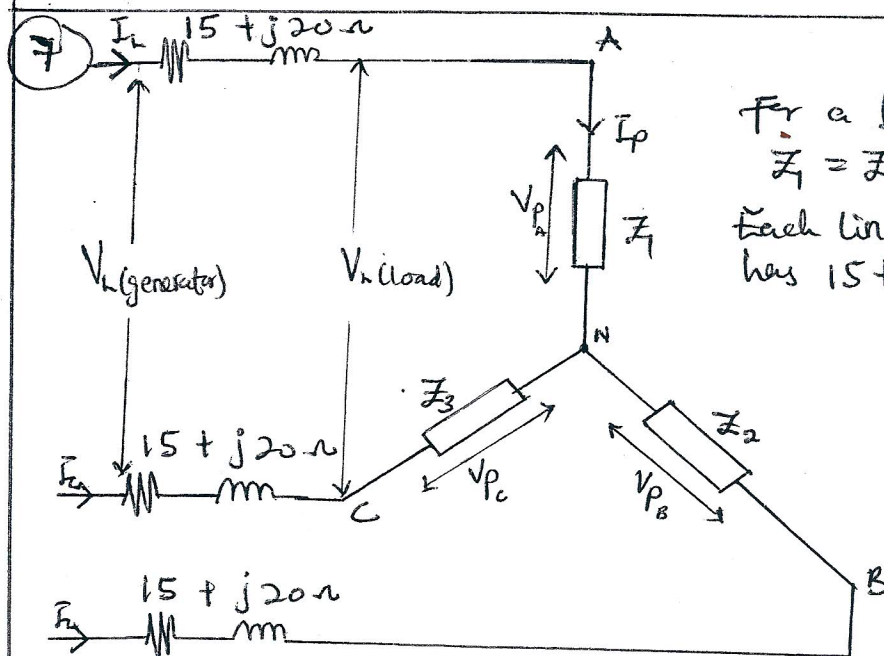
$P = \sqrt{3} \times 250\sqrt{3} \times 30.2 \times \cos 52.85^\circ$

$P = 13674.7972$

$P = 13678.42 \text{ W}$

$\Rightarrow P = 13.67 \text{ kW}$

$\therefore P = \underline{\underline{13.7 \text{ kW}}}$



For a balanced load
 $Z_1 = Z_2 = Z_3$
 Each line transmission
 has $15 + j20$ impedance.

① $V_{L(\text{load})} = 12 \text{ kV}$
 $P_T = 160 \text{ kW}$
 $I_L = I_p$

$\text{Pf} (\cos \phi) = 0.86 \text{ lagging}$

$$P = \sqrt{3} V_L I_L \cos \phi$$

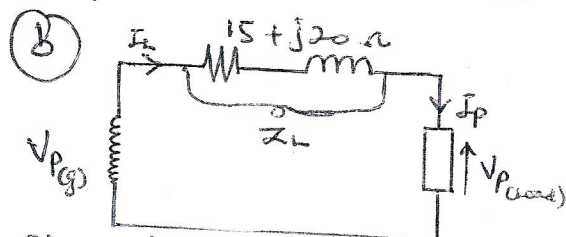
$$I_L = \frac{P}{\sqrt{3} \times V_L \cos \phi}$$

$$I_L = \frac{160000}{\sqrt{3} \times 12000 \times 0.86}$$

$$\therefore I_L = 8.95 \text{ A}$$

$$\phi = \cos^{-1} 0.86 = 30.68^\circ \text{ lagging}$$

$$\Rightarrow I_p = I_L = 8.95 \angle -30.68^\circ$$



Phase Voltage of generator ($V_{p(g)}$)

$$V_{p(g)} = I_p Z_L + V_{p(\text{load})}$$

$$V_{p(g)} = (8.95 \angle -30.68^\circ)(15 + j20) + 6928.20 \angle 0^\circ$$

$$V_{p(g)} = 7135.50 \text{ V} = 7.1 \text{ kV} \angle 0.68^\circ$$

$$\therefore V_{p(g)} = 7.1 \text{ kV}$$

② $30.68 + 0.68$
 $31.36 \approx \underline{\underline{31.4^\circ}}$

$$\begin{aligned}
 (8) \quad V_L &= 450 \text{ V} \\
 f &= 50 \text{ Hz} \\
 S &= 3.75 \text{ kVA} \\
 \text{Pf}(\cos \phi) &= 0.6 \text{ lagging}
 \end{aligned}$$

$$(a) \quad S = V_L I_L \sqrt{3}$$

$$I_L = \frac{S}{\sqrt{3} V_L}$$

$$I_L = \frac{3.75 \times 10^3}{\sqrt{3} \times 450}$$

$$\therefore I_L = \underline{\underline{4.811 \text{ A}}}$$

$$(b) \quad I_L = I_p$$

$$X_L = 2\pi f L$$

$$L = \frac{X_L}{2\pi f}$$

$$Z = \frac{V_p}{I_p} = \frac{259.81}{4.811}$$

$$Z = 54 \Omega$$

$$\cos \phi = 0.6$$

$$\phi = \cos^{-1} 0.6$$

$$\phi = 53.13$$

$$Z = 54 \angle 53.13 = 32.4 + j43.2$$

$$\Rightarrow Z = 54 +$$

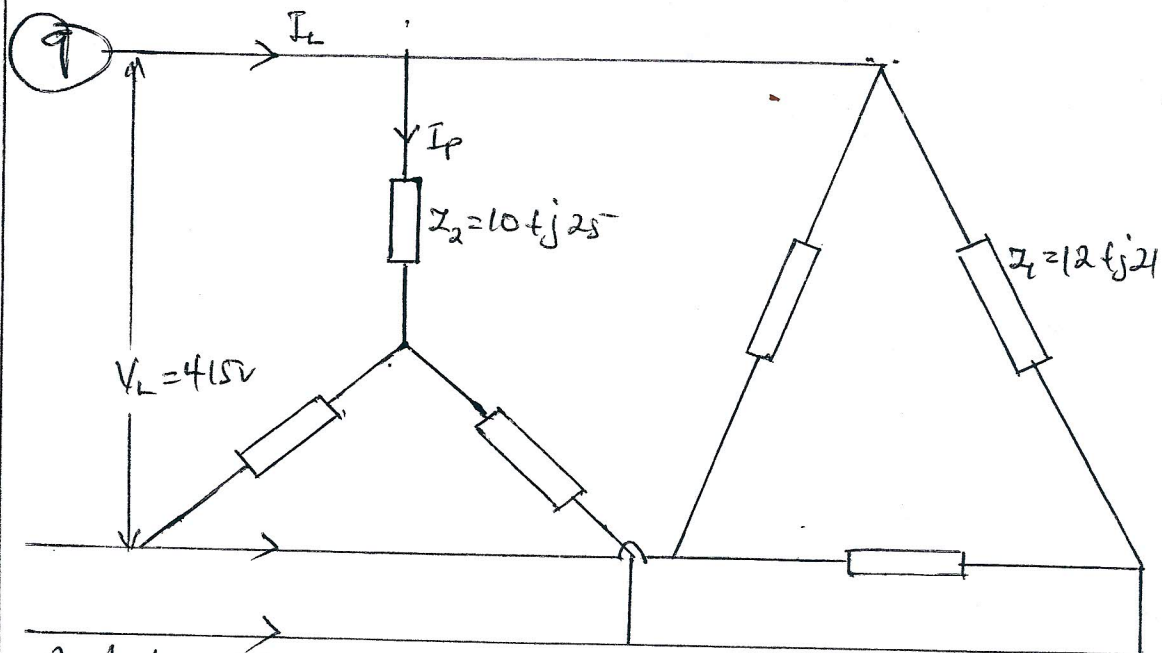
$$\Rightarrow 32$$

$$\Rightarrow Z = 32.4 + j43.2$$

$$\Rightarrow X_L = 43.2$$

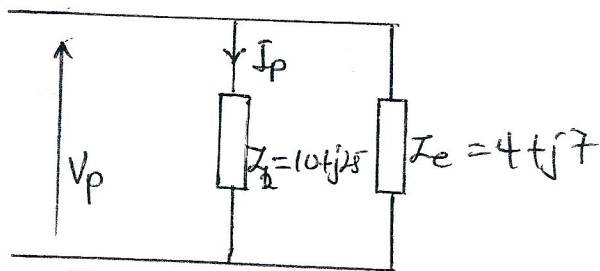
$$\Rightarrow L = \frac{43.2}{2\pi \times 50}$$

$$\therefore L = \underline{\underline{0.138 \text{ H}}}$$



Replacing the delta connected impedance with its star impedance (Z_e)

$$\rightarrow Z_e = \frac{12 + j24}{3} = 4 + j7$$



$$a) V_L = \sqrt{3} V_P$$

$$\Rightarrow V_P = \frac{V_L}{\sqrt{3}} = \frac{415}{\sqrt{3}} = 239.6V$$

$$V_P = I_P Z$$

$$\Rightarrow I_P = \frac{V_P}{Z_2} + \frac{V_P}{Z_1}$$

$$I_P = \frac{239.6 \angle 0^\circ}{10 + j25} + \frac{239.6 \angle 0^\circ}{4 + j7}$$

$$\therefore I_P = 38.551A \angle -62.12^\circ$$

$$b) P.f = \cos \phi = \cos (62.12) = 0.47 \text{ (lagging)}$$

$$c) P_i = 3 V_P I_P \cos \phi$$

$$= 3 \times 239.6 \times 38.551 \times 0.47$$

$$\therefore P_i = 13 \text{ kW}$$

$$(10) V_L = 450V$$

$$I_L = 10A$$

$$P_f (\cos \phi) = 0.85 \text{ leading}$$

$$(a) V_L = V_p = 450V$$

$$V_p = I_p Z$$

$$\Rightarrow Z = \frac{V_p}{I_p}$$

$$I_p = \frac{I_L}{\sqrt{3}} = \frac{10}{\sqrt{3}} = 5.7735A$$

$$\Rightarrow Z = \frac{450}{5.7735}$$

$$\therefore Z = \underline{\underline{77.942 \Omega}}$$

$$(b) X_c = \frac{1}{2\pi f C} \quad \text{--- (1)}$$

$$Z = 77.942$$

$$\cos \phi = 0.85$$

$$\phi = \cos^{-1} 0.85$$

$$\phi = 31.788^\circ$$

$$Z = 77.942 \angle 31.788^\circ$$

$$Z = 66.25 + j41.058$$

$$\Rightarrow X_c = 41.058 \Omega$$

From eqn (1)

$$X_c 2\pi f C = 1$$

$$C = \frac{1}{2\pi f X_c}$$

$$\Rightarrow C = \frac{1}{2\pi \times 50 \times 41.058} = 7.753 \times 10^{-5} F$$

$$C = 77.53 \times 10^{-6} F$$

$$\therefore C = \underline{\underline{77.53 \mu F}}$$

$$(c) P = V_p I_p \cos \phi$$

$$P = 450 \times 5.7735 \times 0.85$$

$$P = 2208.36 W$$

$$\therefore P = \underline{\underline{2.208 kW}}$$

Ref: MATESA KNUST

$$\textcircled{W} \begin{aligned} I_R &= 10 + j5 \\ I_Y &= 15 + j10 \\ I_B &= 8 + j5 \end{aligned}$$

$$V_p = \frac{V_L}{\sqrt{3}} = \frac{415}{\sqrt{3}} = 239.6 \text{ V}$$

$$\textcircled{a} I_{P_1} = \frac{239.6 \angle 0}{15 + j10}$$

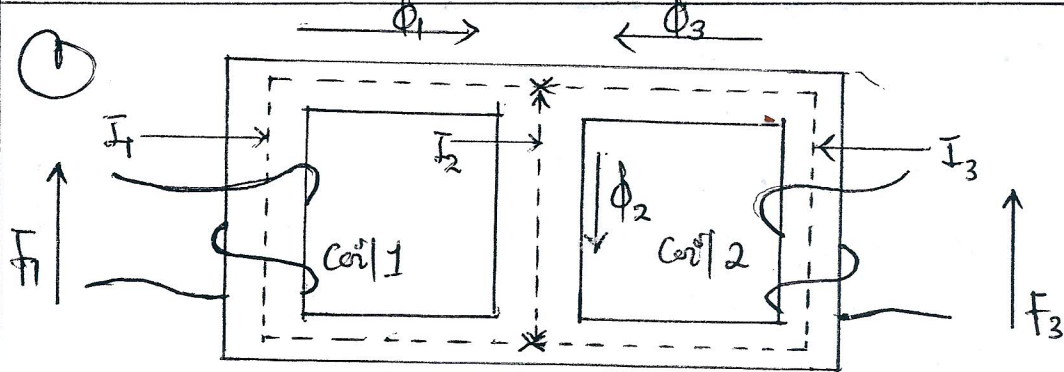
$$\therefore I_{P_1} = \underline{\underline{13.291 \angle -33.69}}$$

$\textcircled{B}$

BY DR. ~~REXIN~~

SOLUTION

SELF ASSESSMENT 5



POWERED BY KUMI GEROGE

Ref: MATESA KNUST

② $I = 9.2 \text{ A}$
 $N = 500$
 $\phi = 0.3 \text{ mWb} = 0.3 \times 10^{-3} \text{ Wb}$
 $\text{mmf} = F = N \times I$
 $= 500 \times 9.2$
 $\therefore \text{mmf} = \underline{4600 \text{ AT}}$

⑥ $F = \phi R$
 $R = \frac{F}{\phi} = \frac{4600}{0.3 \times 10^{-3}}$
 $\therefore R = \underline{1.533 \times 10^7 \text{ AT/Wb}}$

③ $N = 600$
 $I = 0.2 \text{ A}$
 $\phi = 200 \text{ mWb} = 200 \times 10^{-3} \text{ Wb}$
 $\text{mmf} = F = N I$
 $= 600 \times 0.2$
 $\therefore \text{mmf} = \underline{120 \text{ AT}}$

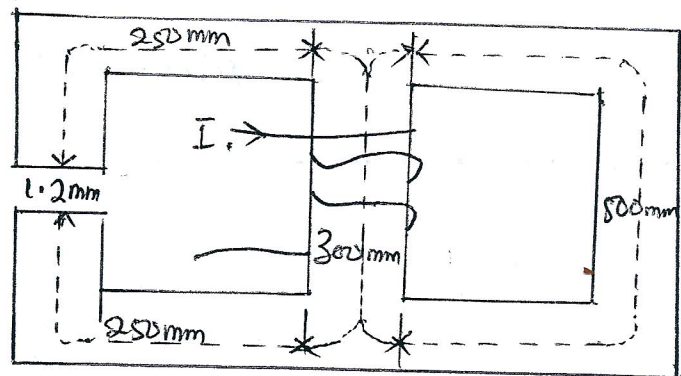
⑥ $R = \frac{F}{\phi} = \frac{120}{200 \times 10^{-3}}$
 $\therefore R = \underline{600 \text{ AT/Wb}}$

④ $N = 3000$
 $A_w = 0.05 \text{ m}^2$
 $l_w = 2 \text{ m}$
 $\phi = 10 \mu\text{Wb} = 10 \times 10^{-6} \text{ Wb}$
 $F = NI$
 $F = \phi R$
 $F = \frac{\phi l_w}{\mu_0 \mu_r A}$
 $F = \frac{10 \times 10^{-6} \times 2}{4\pi \times 10^{-7} \times 1 \times 0.05}$

$$\Rightarrow I = \frac{F}{N} = \frac{318.3099}{3000}$$

$$\therefore I = \underline{0.106 \text{ A}}$$

(5)



POWERED BY KUMI GEORGE

②

27

A vertical column of hand-drawn sketches in blue ink. From top to bottom, the sketches include: a small triangle with internal lines; a series of overlapping loops and lines; a heart-like shape formed by two overlapping loops; and a large, open, triangular shape at the bottom.

⑦ $A = 0.5 \text{ m}^2$
 $\text{mmf} = F = 1000 \text{ A/m}$
 $\phi = ? \quad \mu_r = 500 \quad L = 1 \text{ m}$
 $F = \phi R$

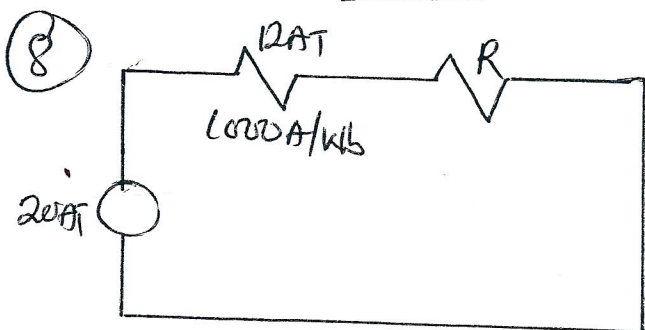
$\phi = \frac{F}{R}$ where $R = \frac{L}{\mu_0 \mu_r A}$

$\Rightarrow R = \frac{1}{4\pi \times 10^{-7} \times 500 \times 0.5} = 3183.09886$

$\phi = \frac{1000}{3183.09886} = 0.314159$

$\Rightarrow \phi = 314.159 \times 10^{-3} \text{ Wb}$

$\therefore \phi = \underline{\underline{314.159 \text{ mWb}}}$



$20 = 12 + R\phi$

$\Rightarrow \text{Flux through } R (R\phi) = 8 \text{ AT}$

If $12 \text{ AT} \rightarrow 1000 \text{ A/Wb}$
 then $8 \text{ AT} \rightarrow R$

$R = \frac{8}{12} \times 1000$

$\therefore R = \underline{\underline{666.667 \text{ A/Wb}}}$

DR-BRAXIN
BY

$$(10) N = 3000$$

$$I = 0.5 \text{ A}$$

$$A = \pi r^2; r = 2 \text{ cm} = 2 \times 10^{-2} \text{ m}$$

$$A = \pi \times (2 \times 10^{-2})^2 \text{ m}^2$$

$$A = 1.2566 \times 10^{-3} \text{ m}^2$$

$$L = 2\pi r; r = 50 \text{ cm} = 50 \times 10^{-2} \text{ m}$$

$$L = 2\pi \times 50 \times 10^{-2} \text{ m}$$

$$L = \pi = 3.14159 \text{ m}$$

$$F = \phi R$$

$$\text{but } F = NI = 3000 \times 0.5 \\ = 1500 \text{ AT}$$

$$R = \frac{L}{\mu_0 \mu_r A} = \frac{3.14159}{4\pi \times 10^{-7} \times 1 \times 1.2566 \times 10^{-3}}$$

$$\Rightarrow R = 19.89 \times 10^8$$

$$(a) B = \frac{\phi}{A}$$

$$\text{where } \phi = \frac{F}{R} = \frac{1500}{19.98 \times 10^8}$$

$$\phi = 7.5415 \times 10^{-7} = 0.754 \times 10^{-6} \text{ Wb}$$

$$\Rightarrow B = \frac{0.754 \times 10^{-6}}{1.2566 \times 10^{-3}} \\ = 6.0 \times 10^{-4} \text{ Wb/m}^2 \\ = 0.6 \times 10^{-3} \text{ T}$$

$$\therefore B = \underline{\underline{0.6 \text{ mT}}}$$

$$(b) \phi = \frac{F}{R}$$

$$= \frac{1500}{19.98 \times 10^8} = 7.5415 \times 10^{-7}$$

$$\Rightarrow \phi = 0.754 \times 10^{-6} \text{ Wb}$$

$$\therefore \phi = \underline{\underline{0.754 \mu\text{Wb}}}$$

XXOTIVATIONAL XMESSAGES

- ① The world will only seek first for the man who knows. The devil and the world will only bow to the man or woman who knows.
Knowledge is power.
- ② Those who keep their daily appointments with God don't get disappointments from men.
- ③ Nobody can keep you down, unless you decide not to rise again.
- ④ When you get a second change in life, do not regret, hold it with both hands and make things happen.
- ⑤ Nothing is impossible untill you decide to stop trying. The Secret of Success is to start from the scratch and keep on scratching.
- ⑥ Until you enjoy what you are doing, it becomes afflictions.
- ⑦ Hardwork and great joy in what you ~~are~~ are doing are two powerful success keys.
- ⑧ Hardwork is important to Success but those that despise their small beginnings never live to see big endings.
- ⑨ If you ~~do~~ always do what you always do, you will always get what you always get.
- ⑩ You must determine to be an example and a first partaker in all that you lay your hand to do or preach.
- ⑪ Never forget that if you die without fulfilling your purpose in life, death is not gain but pain.
- ⑫ The minute you settle for less than you deserve you ~~are~~

BY D R - B R A I A